Data Management

- Data is a valuable resource of the organization
- Data management is a key part of the business processes within the organization

Data Are Resources of the Organization

Data are valuable assets of the organization

There are many types of data:

- Master data (e.g., customers)
- Transaction data (e.g., sales)
- Historical data (e.g., credit history)
- Secondary data (e.g., industrial average)
- Subjective data (e.g., marketing survey)

Data, Information, Knowledge

- Data are raw facts
  - The nature of data is raw and without context
- Information is a product of processed data in a given context
  - Information is data that have been given a semantic meaning
- Knowledge is human interpretation of the real world
  - Knowledge guides actions

Data, Information, Knowledge

- Database
  - Store data
    - Not information or knowledge
  - Generate information
- Database users can use the generated information to develop knowledge
- Data warehouses and knowledge bases are built on databases, and they deal with information and knowledge

Control Data Redundancy

- The central issue of database is data redundancy control
- **Data redundancy** occurs when the same **fact** is stored in more than one place
  - Note: Do not interpret data redundancy to be "the same number or the same text is stored in more than one place."

Example of Data Redundancy and Data Integration

CUSTOMER- PURCHASE

CustomerNumber	CustomerName	CustomerAddress	CustomerPhone	Purchase	PurchaseTime
123456	Smith	Westport Rd.	508999	$200	10/20/2014 09:45am
234562	Green	Eastport Ave.	509343	$300	10/23/2014 01:32pm
123456	Smith	Westport Rd.	508999	$150	11/08/2014 05:04pm

(a) Data redundancy occurs

Example of Data Redundancy and Data Integration (cont'd)

CUSTOMER

CustomerNumber	CustomerName	Customer Address	CustomerPhone
123456	Smith	Westport Rd.	508999
234562	Green	Eastport Ave.	509343

PURCHASE

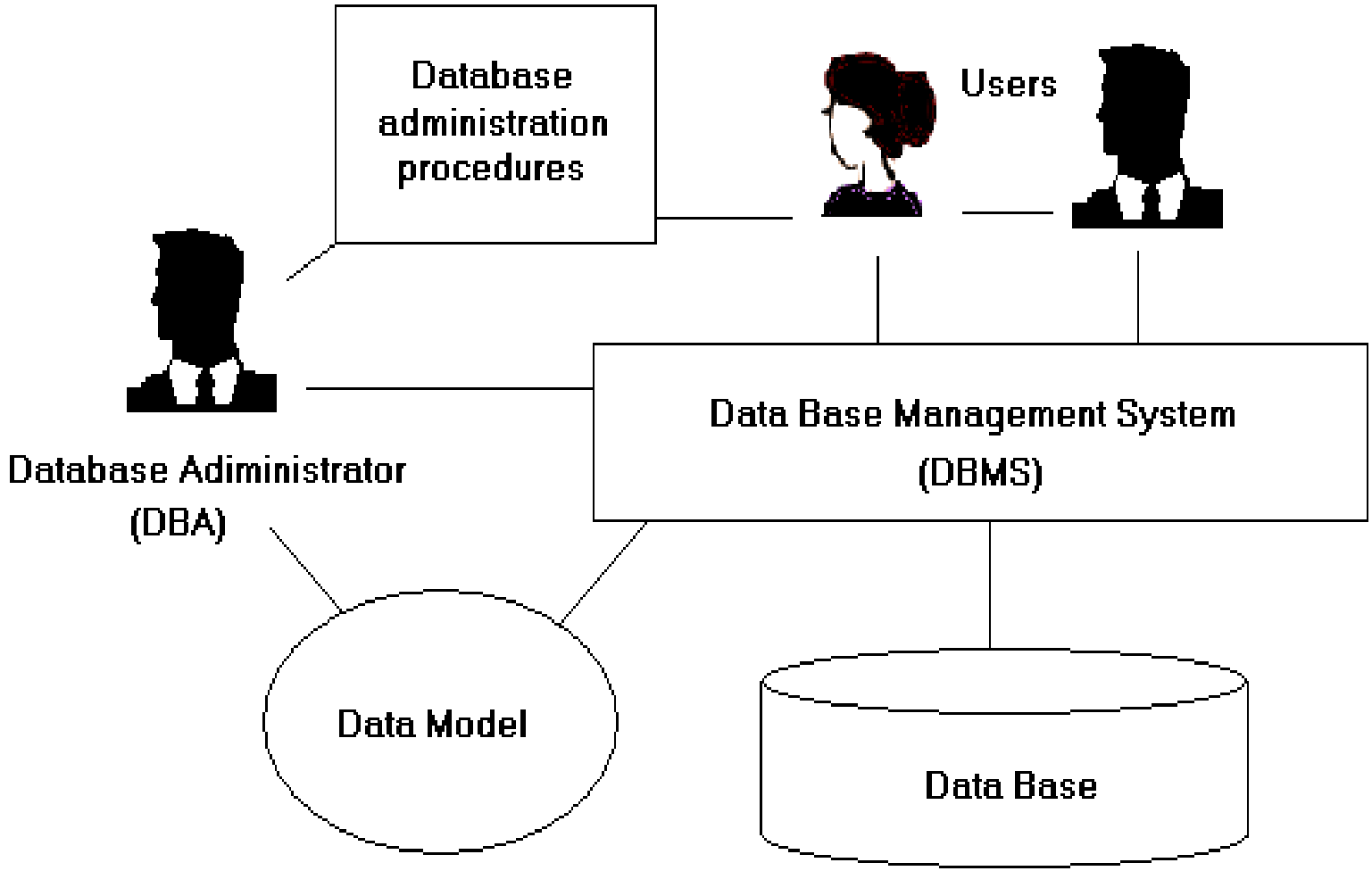
CustomerNumber	Purchase	PurchaseTime
123456	$200	10/20/2014 09:45am
234562	$300	10/23/2014 01:32pm
123456	$150	11/08/2014 05:04pm

(b) There is no data redundancy if the data are stored in this form

Database and Database Systems

- Objectives of database systems
  - Controlled data redundancy
  - Data consistency
  - Data sharing
  - Facilitate application development
  - Wide-ranging data management functions

Database System



Database Management System (DMBS)

- DBMS is software that manages databases
  - Support database construction and data access
  - Control data redundancy
  - Provide data integration
  - Maintain data independency
  - Monitor and improve data retrieval performance
  - Control data security
  - Enforce business rules and maintain data integrity
  - Manage concurrency control
  - Perform backup and recovery
  - Maintain data dictionary
  - Facilitate restructuring database

Popular DBMS

- (1) ORACLE is Oracle Corporation’s product
- (2) IBM DB2 is IBM’s product
- (3) Microsoft SQL Server is Microsoft’s product
- (3) MySQL is a popular choice of database for use in Web applications, because it is closely tied to the popularity of PHP, an open-source server-side programming language

Data Models

- A data model is a representation of complex data structures of the target real world, or system (e.g., Student Information System), for database design
- There are four data models in the DB field:

  - Hierarchical model for hierarchical databases

  - Network model for network databases

  - Entity-relationship (ER) model for relational databases

  - Object-oriented model for object-oriented databases

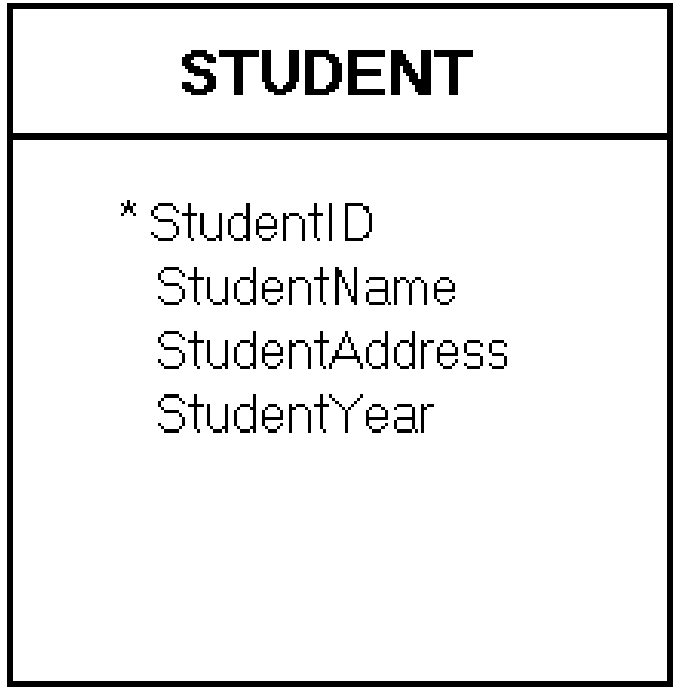
ER Model: Entity

- Entity – a class objects
  - Physiomorphic entity is a physically existing object; e.g., customer, student, and inventory
  - Event entity represents an event of routine operations; e.g., game and credit approval
  - Document entity is an artificial abstraction, e.g., degree and invoice

Entity – e.g., STUDENT

- Attributes – describe the entity, e.g., StudentID, StudentName ….
- Data type – each attribute has its data type, e.g., text, number…
- Instance – a value of attribute, e.g., John Smith
- Primary key – unique identifier of the entity, e.g., StudentID

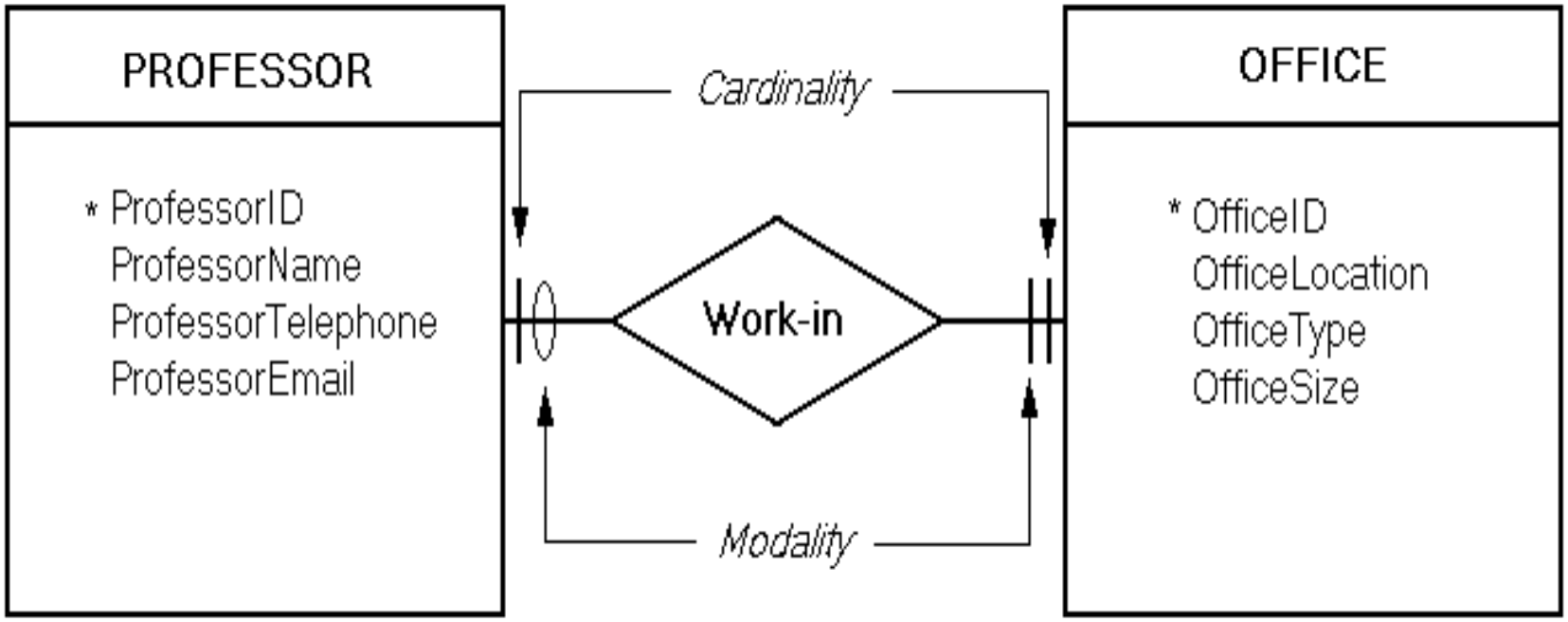
Entity


STUDENT
* StudentID
StudentName
StudentAddress
StudentYear

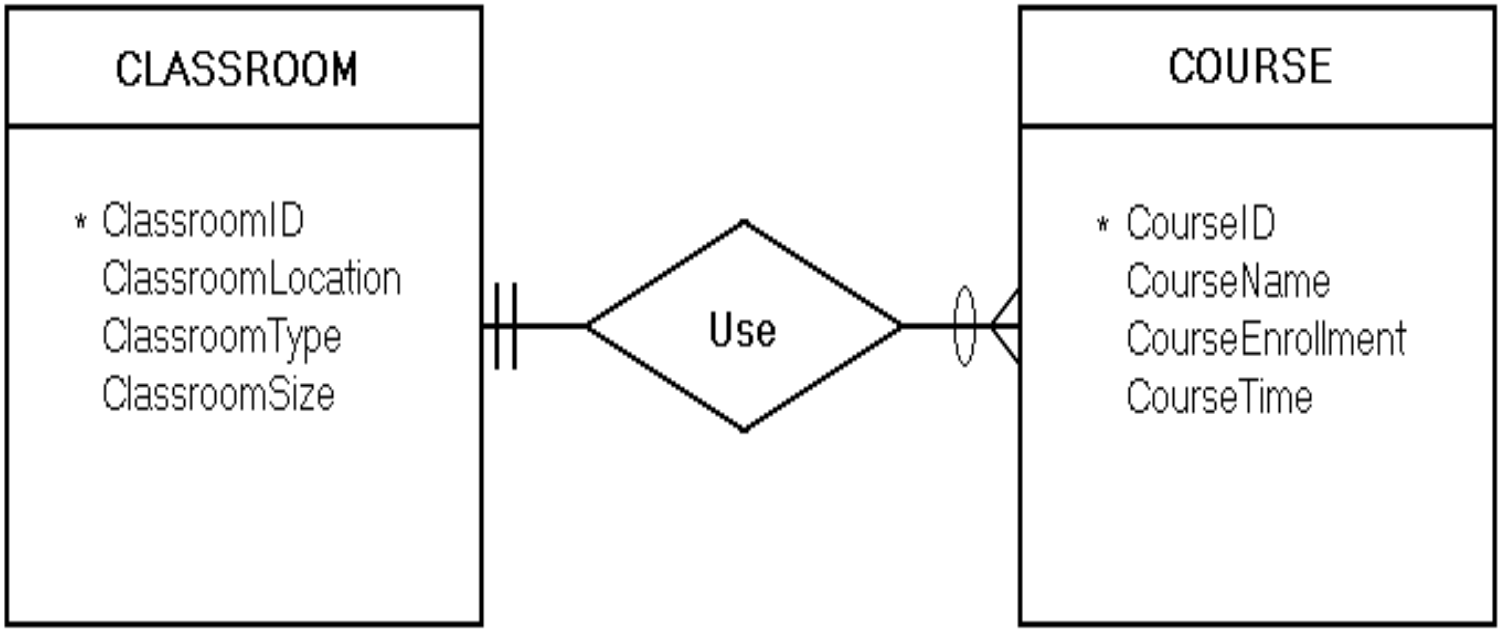
ER Model: Relationship

- Relationship – Association between entities
  - Binary
  - Unary
  - Ternary
- Cardinality - Maximum number of instances of entities involved in the relationship
- Modality - Minimum number of instances of entities involved in the relationship

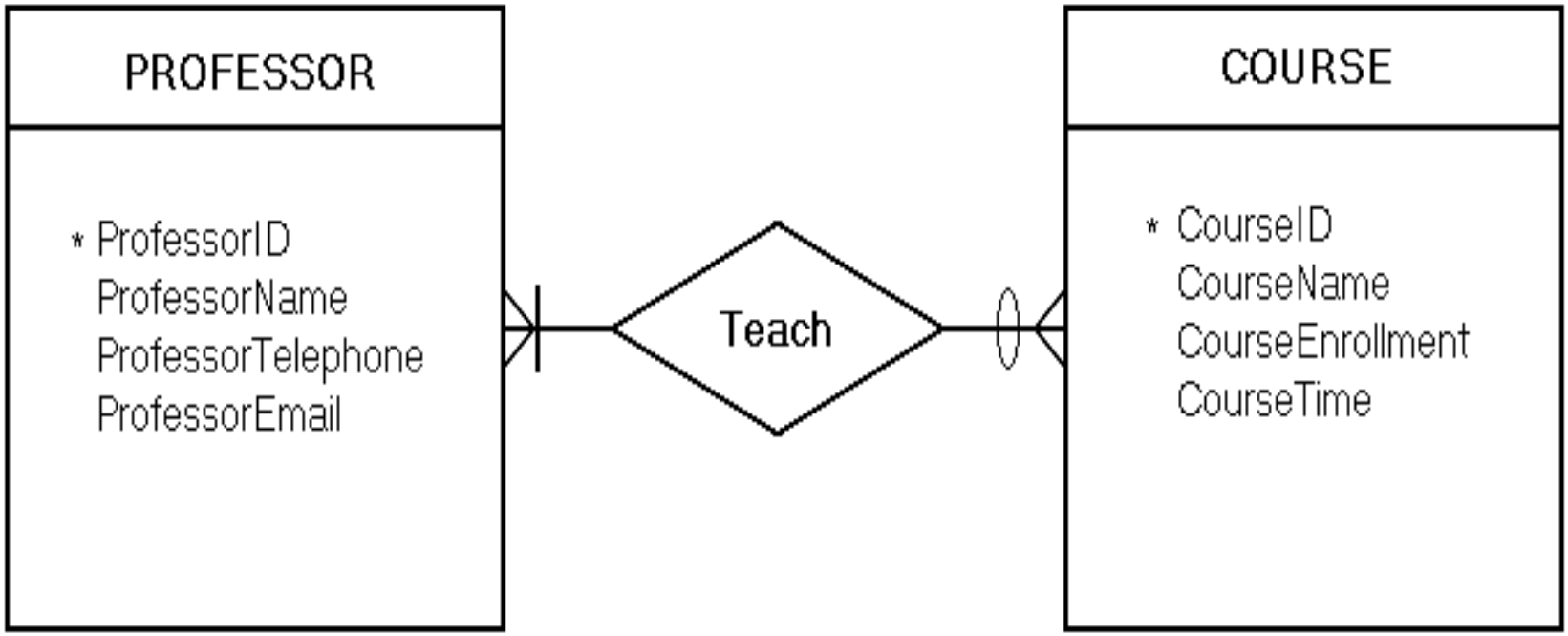
Binary, 1:1


PROFESSOR
* ProfessorID
ProfessorName
ProfessorTelephone
ProfessorEmail
Cardinality
Work-in
Modality
OFFICE
* OfficeID
OfficeLocation
OfficeType
OfficeSize

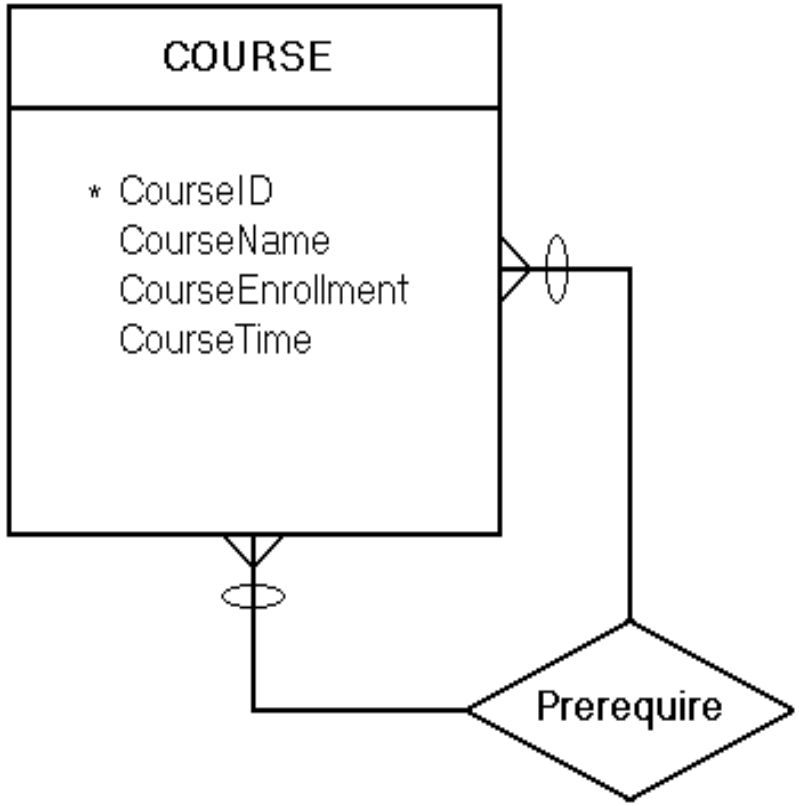
Binary, 1:M


CLASSROOM
* ClassroomID
ClassroomLocation
ClassroomType
ClassroomSize
Use
COURSE
* CourseID
CourseName
CourseEnrollment
CourseTime

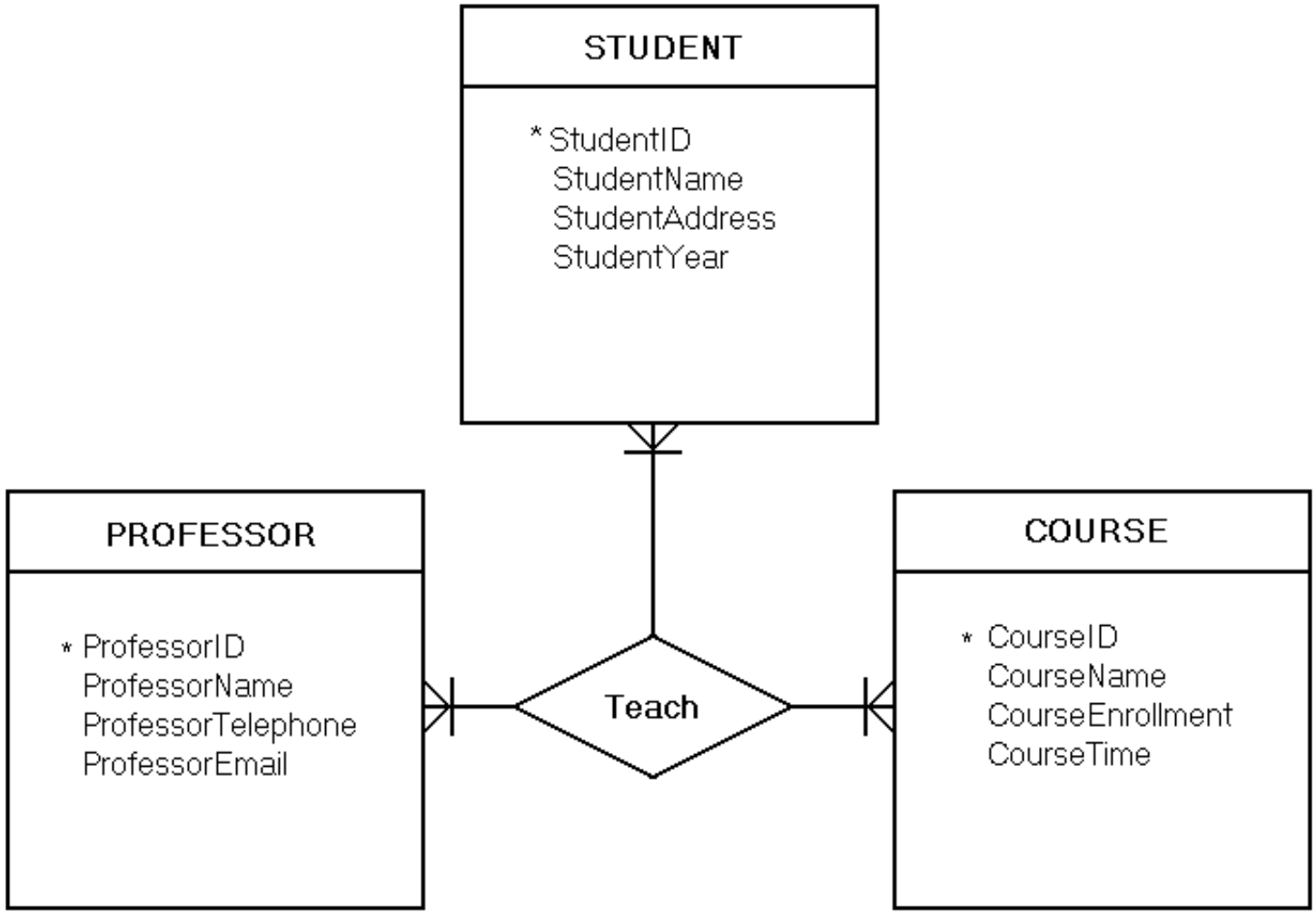
Binary, M:M


PROFESSOR
* ProfessorID
ProfessorName
ProfessorTelephone
ProfessorEmail
Teach
COURSE
* CourseID
CourseName
CourseEnrollment
CourseTime

Unary, M:M


COURSE
* CourseID
CourseName
CourseEnrollment
CourseTime
Prerequire

Ternary, M:M


STUDENT
* StudentID
StudentName
StudentAddress
StudentYear
PROFESSOR
* ProfessorID
ProfessorName
ProfessorTelephone
ProfessorEmail
Teach
COURSE
* CourseID
CourseName
CourseEnrollment
CourseTime

ER Diagram

- ER diagram – a graphical tool for ER model
- There is no standard ER diagram notations, and each company uses its own “standard”
- For professionals, it does not matter what “standard” of ER diagram is used, because they know exactly what a notation means
- For beginners, ER diagrams could be confusing
- We will use our “standard” physical ER diagram in UML

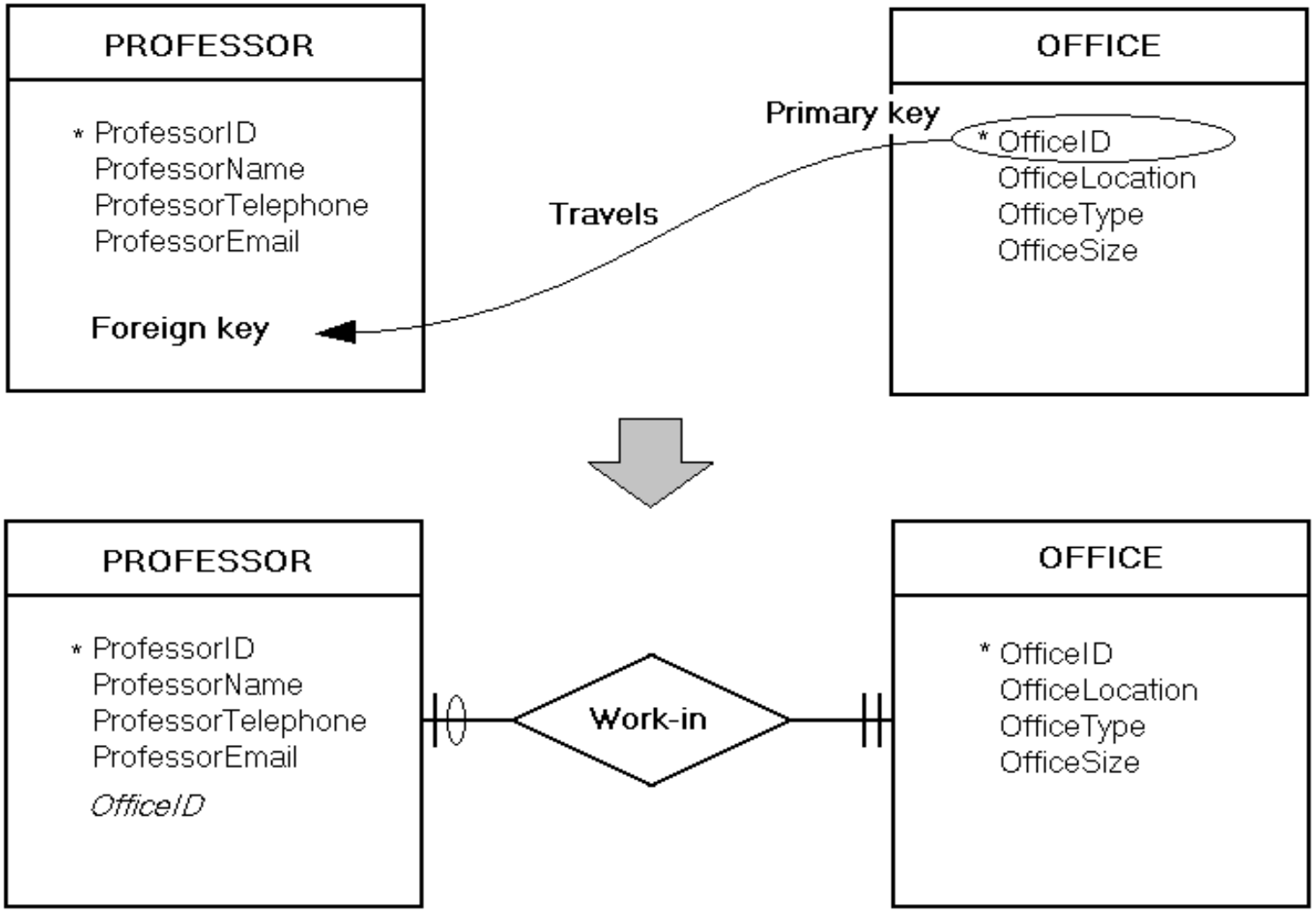
ER Model

- (1) The relationships between the instances of entities along with the cardinalities and modalities are not given, but are modeled by the database designer using ER diagrams. They depend on the **business rules** of the business environment, or the **assumptions** made for the particular database
- (2) The cardinalities and modalities will be used by the database management system to enforce the pertinent business rules
- (3) In relational database, there is no way to present M:M relationships directly in the database. Any M:M relationship must be converted into 1:M relationships through the use of **associative entity**

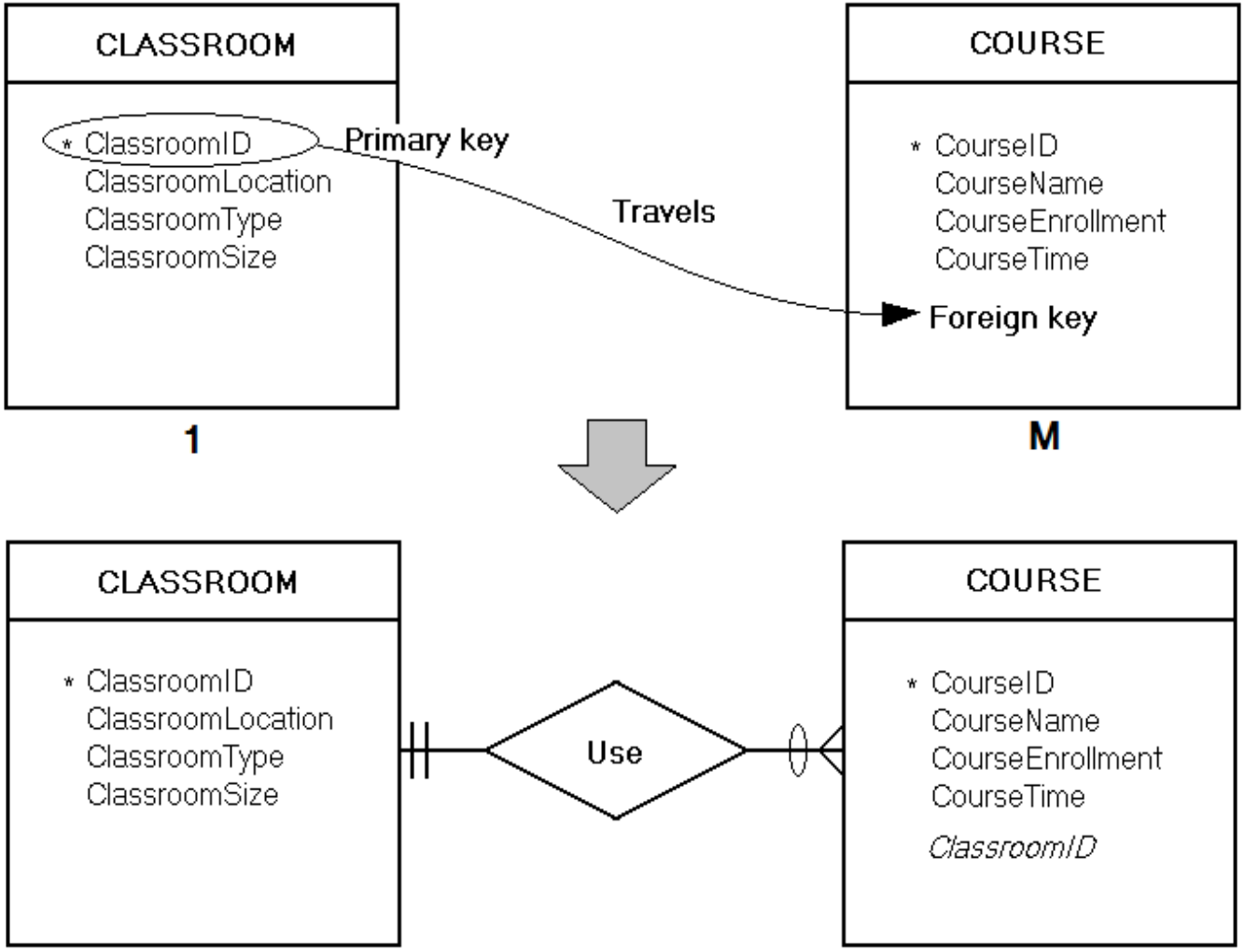
Instrument for Implementing 1:1 and 1:M Relationships – Foreign Key

- For a 1:1 relationship, you can arbitrarily choose the primary key from either entity and place it as the foreign key to the other entity However, if one of the modalities is 0, it is better to place the foreign key into the entity with the 0 modality
- For a 1:M relationship, you place the primary key of the entity with the 1 cardinality into the entity with the M cardinality as the foreign key
- Foreign key in a table and its corresponding primary key in another table have the same domain of values

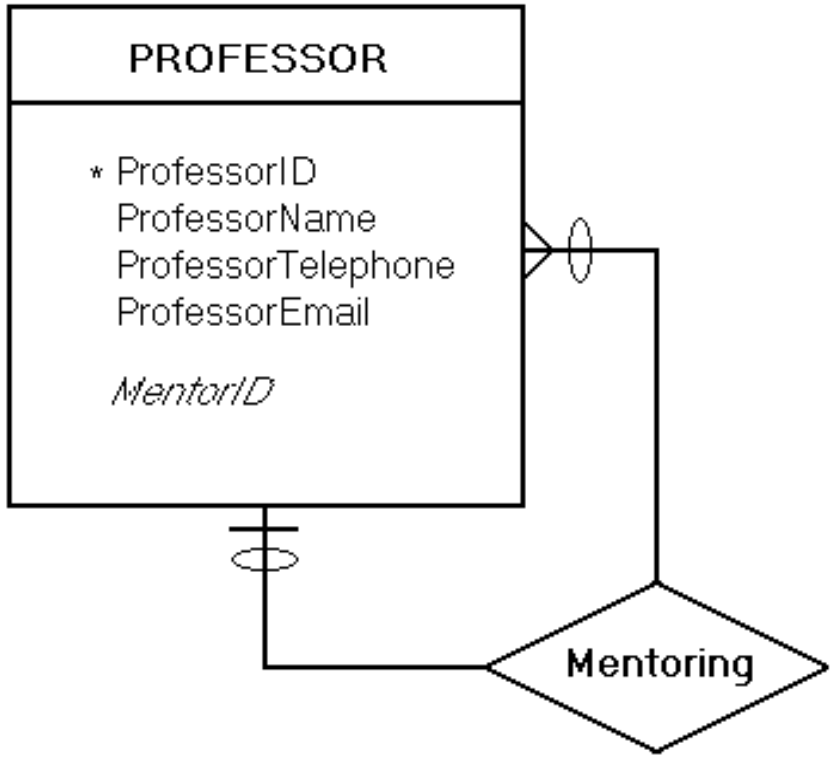
Foreign Key for 1:1



Foreign key for 1:M



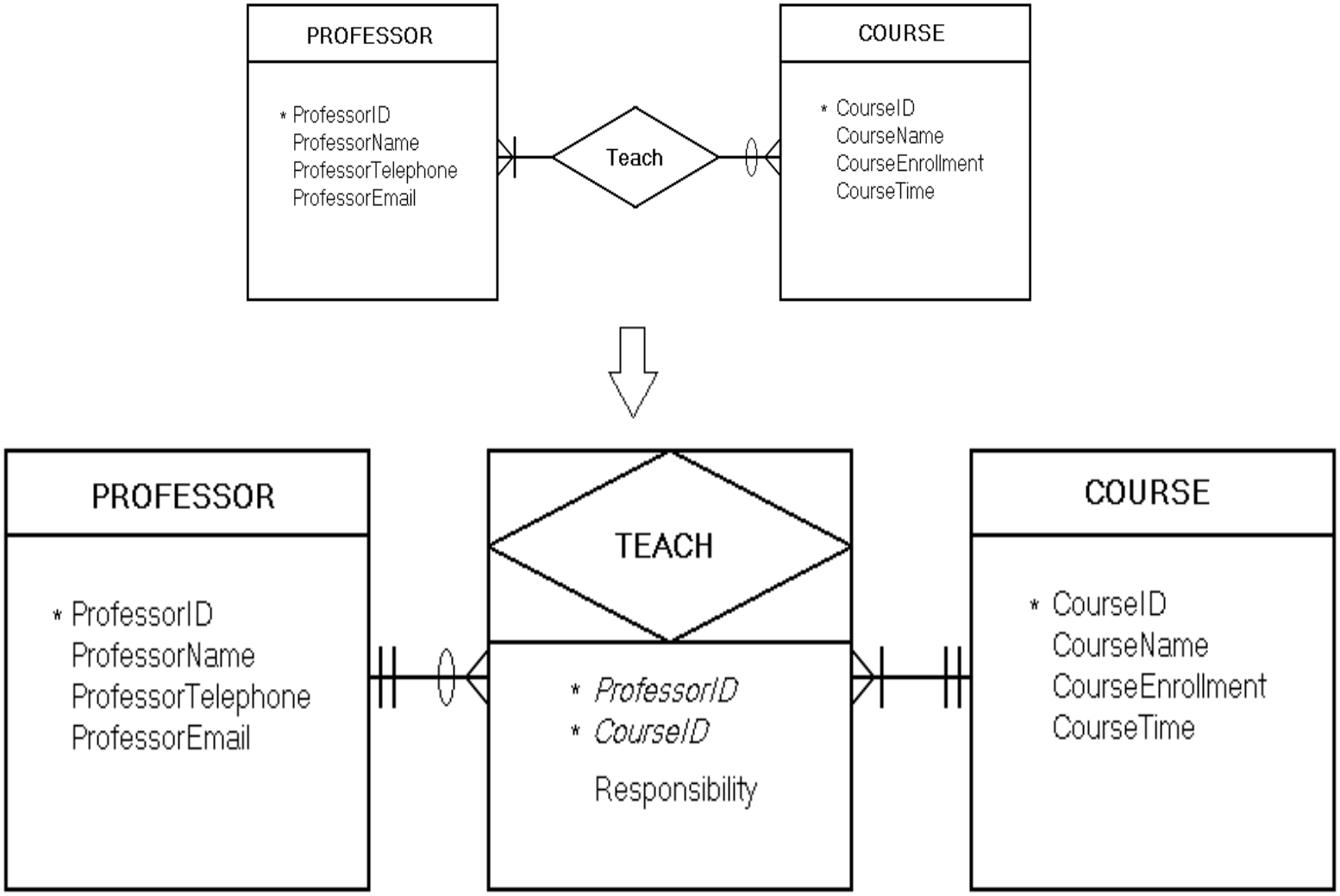
Foreign key in unary 1:M



Instrument for Implementing M:M Relationships – Associative Entity

- Foreign key only is incapable to implement an M:M relationship
- The associative entity converts the M:M relationship into two 1:M relationships which can then be represented in the relational database

Associative entity for implementing M:M relationship



Rules of Associative Entity

- (1) Once an associative entity is introduced, the M:M relationship becomes two 1:M relationships, and the associative entity is on the M sides between the two original entities. The original M:M relationship disappears

- (2) To distinct an associative entity from an ordinary entity, a diamond symbol is used

Rules of Associative Entity (cont'd)

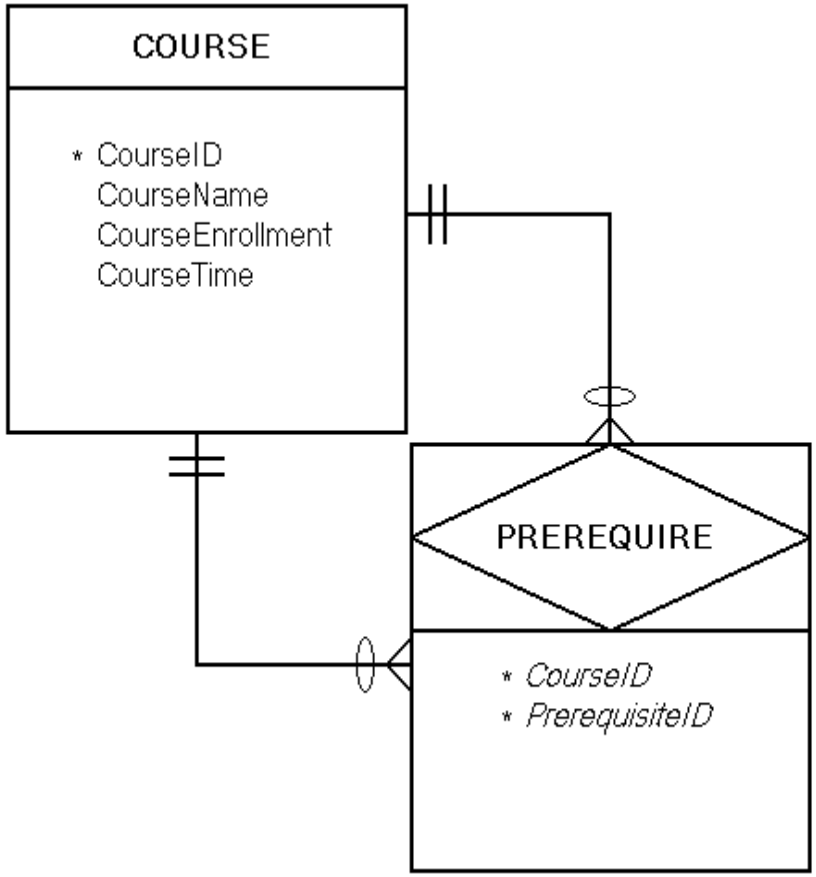
- (3) Follow the same rules of implementing 1:M relationships, place the primary key of the entity on the 1 side into the associative entity as the foreign key

- (4) An associative entity usually has its unique attribute(s) that is not a part of either of the original entity but describes the characteristics of the associative entity, called **intersection data**

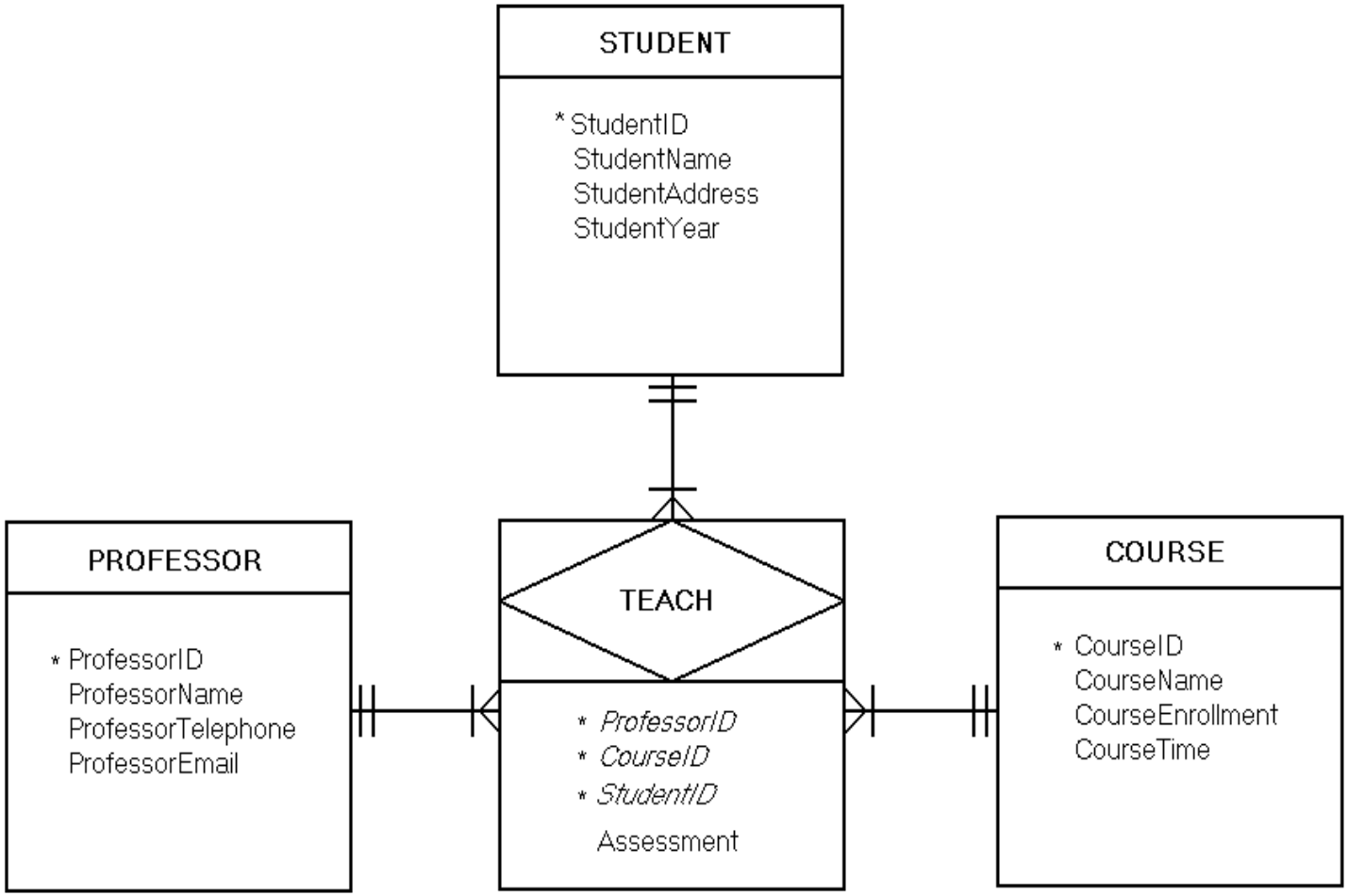
- (5) Usually, the combination of the foreign keys in the associative entity is the primary key of the associative entity

  However, one may add additional attribute(s) to the primary key, or use a new primary key for the associative entity

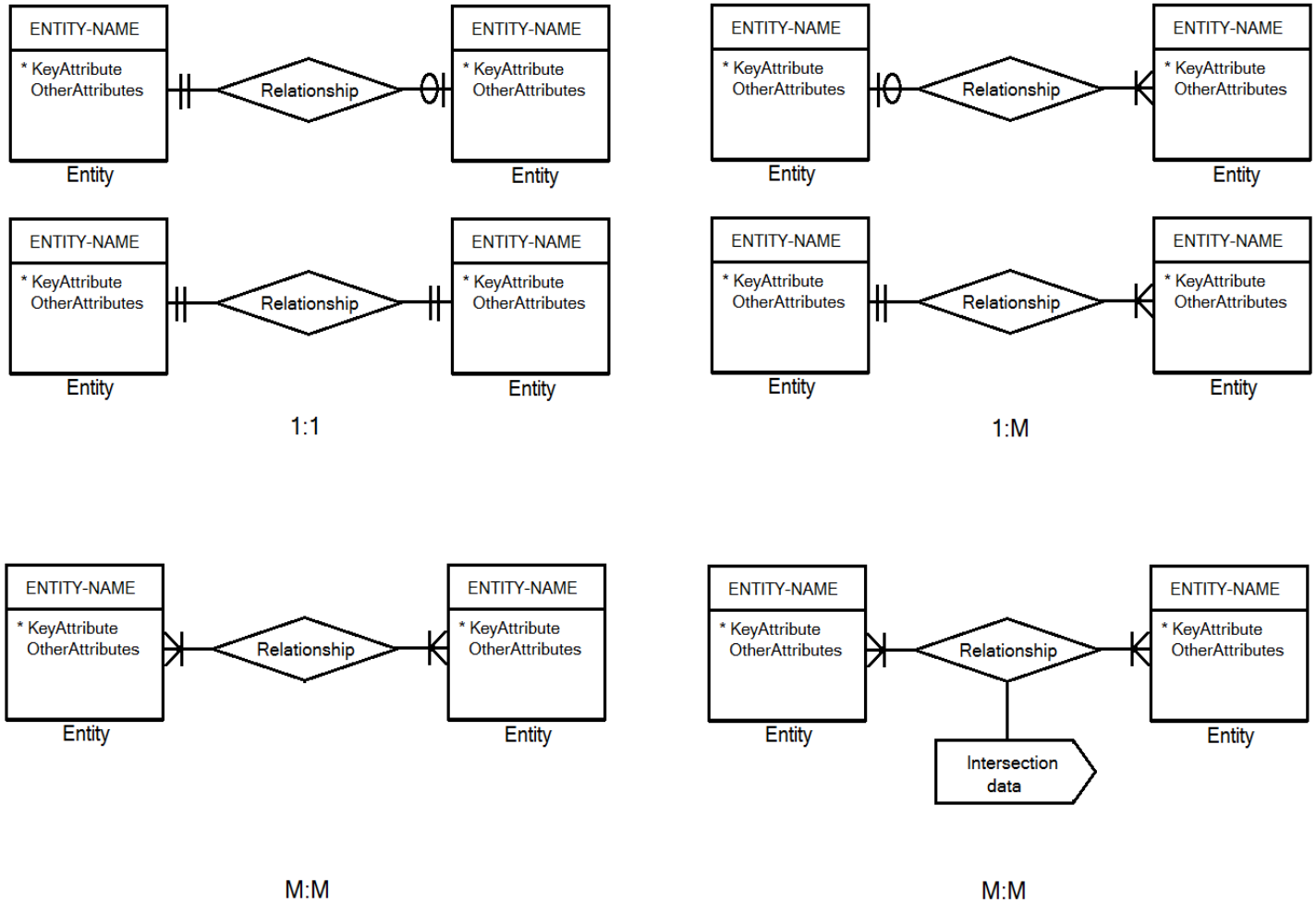
Associative entity in unary



Associative entity in ternary

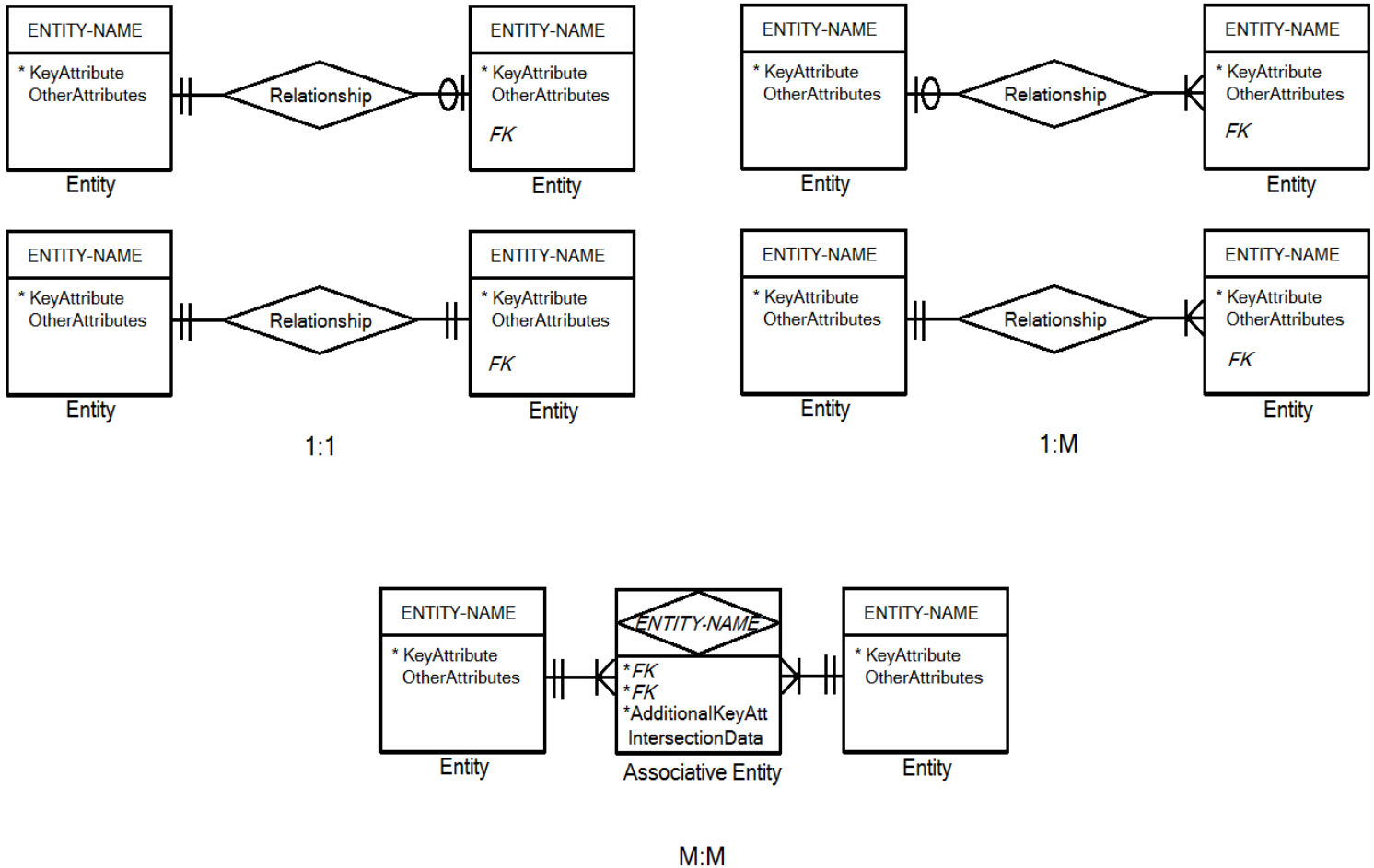


Summary of ER Diagram Notations



(a) Logical ER Diagram Notations (Intermediate Tools)

Summary of ER Diagram Notations (Cont'd)



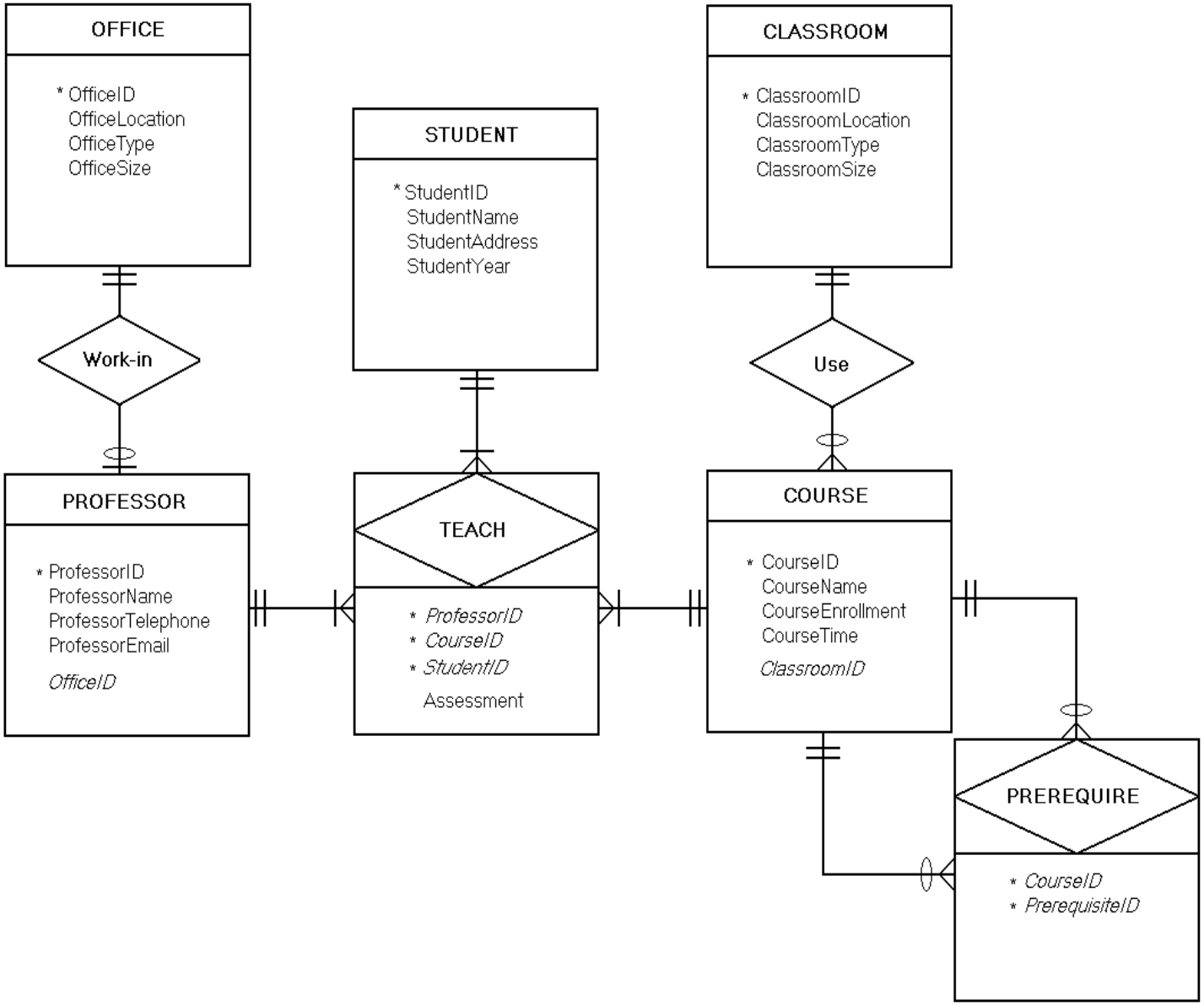
(b) Physical ER Diagram Notations

Start Point for ER Diagram: Transcript

- Transcript of the system (next slide)
- (1) Identify entity and its attributes through identifying **noun**
- (2) Identify relationship through identifying **verb** or verb phrase
- (3) Identify cardinalities and modalities through identifying **business rules**
- (4) Make commonsensical **assumptions** if the transcript does not provide all information you need
- (5) Construct a logical ERD for the case
- (6) If there is an M:M relationship in your logical ERD, convert the M:M relationship into 1:M relationships by adding an associative entity along with its intersection data attributes
- (7) Complete the notations for all primary keys and foreign keys

Example of Transcript

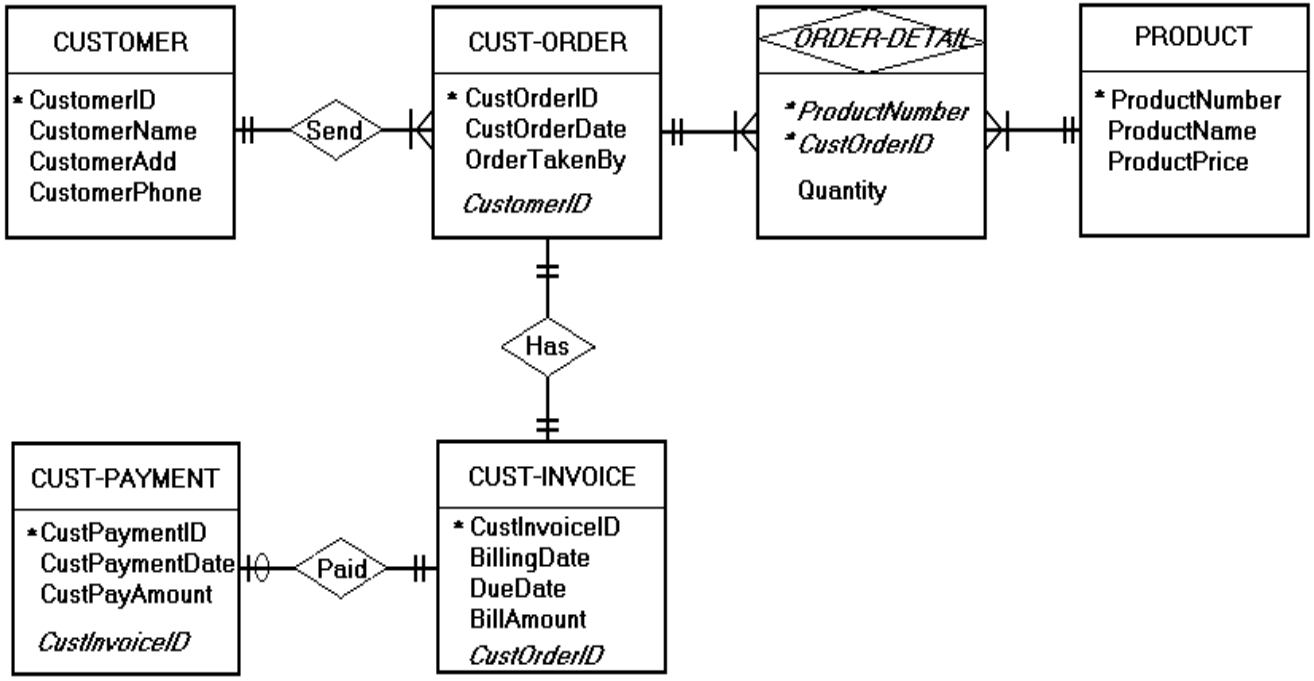
The School wants to create a database to maintain data for the school. Student data should include identification, name, address, and the earliest enrollment year. Professor data should include identification, name phone number, and office. Each professor has her/his own office, and teaches many courses each semester. Several professors might form a team to teach a single course. Each student can take many courses each semester. Course data should include unique course number, title, and enrollment cap. Each course can have its prerequisites. Each course must have a classroom. The school wants to keep track of students' grades.


OFFICE
* OfficeID
OfficeLocation
OfficeType
OfficeSize
STUDENT
* StudentID
StudentName
StudentAddress
StudentYear
CLASSROOM
* ClassroomID
ClassroomLocation
ClassroomType
ClassroomSize
Work-in
Use
PROFESSOR
* ProfessorID
ProfessorName
ProfessorTelephone
ProfessorEmail
OfficeID
TEACH
* ProfessorID
* CourseID
* StudentID
Assessment
COURSE
* CourseID
CourseName
CourseEnrollment
CourseTime
ClassroomID
PREREQUIRE
* CourseID
* PrerequisiteID

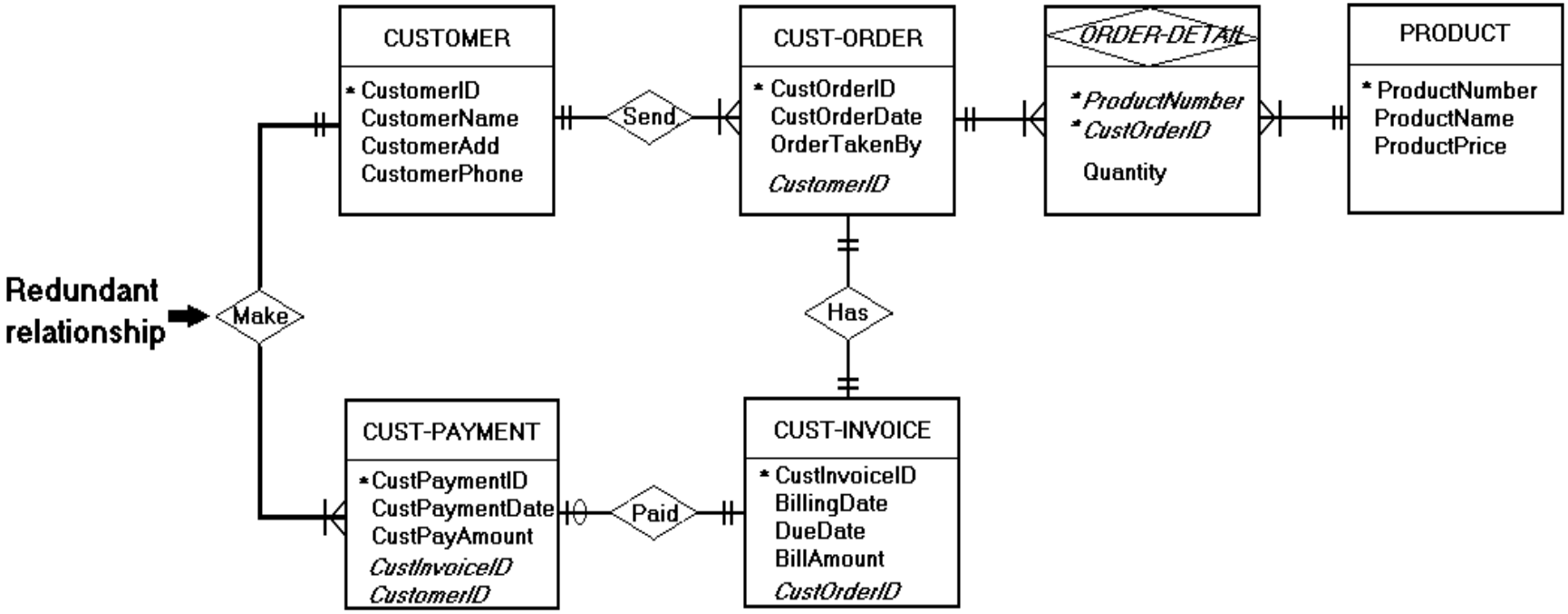
Redundant relationships in ERD

- A redundant relationship in ERD is caused by a **relationship loop**
- However, a relationship loop not always introduces a redundant relationship
- If a relationship loop introduces a redundant relationship, the loop must be broken up by deleting the redundant relationship

An ERD without Redundant Relationship



An ERD with Redundant Relationship



Relational Database Model: Table

- In the relational data model, an entity is a relation, or a table

Attributes

Key

* StudentID	StudentName	StudentAddress	StudentYear
1201	Chris Smith	23 William St.	2016
1334	Anne Riley	38 Union Ave.	2017
1432	John Kerry	17 Wood Lane	2016
2874	Doug Dale	6 Lake Rd.	2018
⋮	⋮	⋮	⋮

Row (Tuple)

Value of the attribute

Properties of Relation

- (1) The order of columns of a relation is not important
- (2) The order of row of a relation is not important
- (3) Each **cell** in the table can have only one single value. The value could be "**null**"
- (4) No two records in a relation have the same value of primary key

Candidate Key and Alternative Key

- A relation can have more than one attribute (or a combination of attributes) that can be used to identify the unique record in the table
- For instance, if we assume StudentName (including first name, middle name, and last name) is unique in the database, then the STUDENT relation can have two **candidate keys**: StudentID and StudentName. If the DBA chooses StudentID to be the primary key for STUDENT, then StudentName is called the **alternative key**

Conversion of ER Model to Relational Database Model

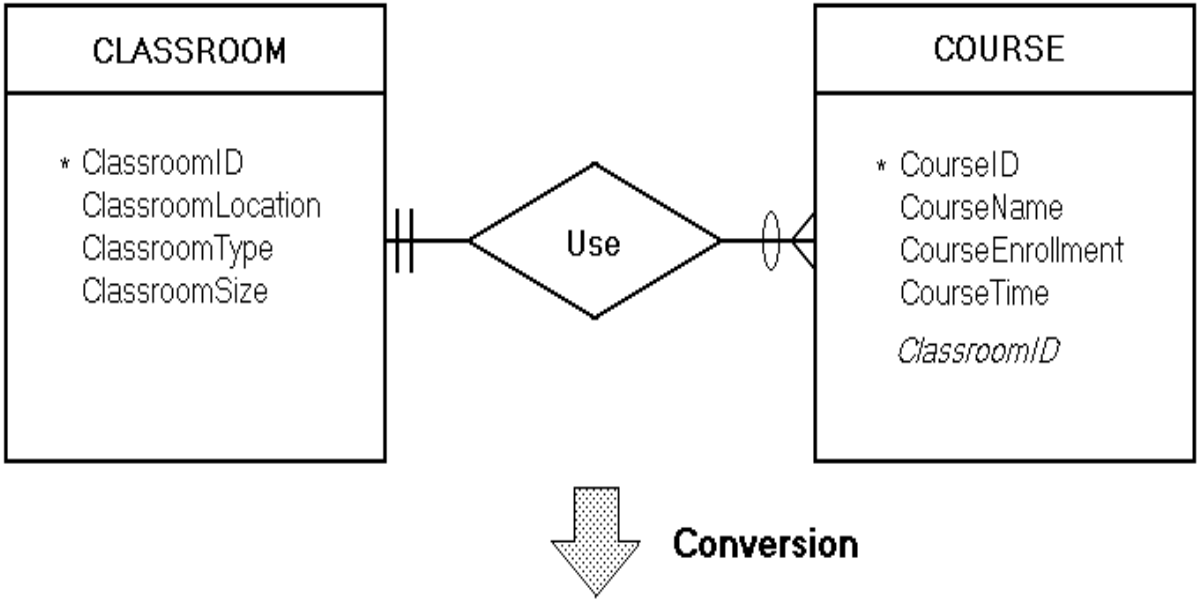
(1)Constructing Tables

(2)Defining Constraints

Constructing Tables

- It is a one seamless step procedure if our "standard" physical ER diagram is used
- (1) An entity in the ER model is a relation in the relational data model
- (2) The primary key of an entity in the ER model is the primary key of the corresponding relation in the relational data model
- (3) The foreign key in an entity in the ER model is an attribute of the corresponding relation in the relational data model

Convert ERD to Relations



CLASSROOM Relation			
* ClassroomID	ClassroomLocation	ClassroomType	ClassroomSize

COURSE Relation				
* CourseID	CourseName	CourseEnrollment	CourseTime	*ClassroomID*

Defining Constraints

- (1) For a relationship with 1:M cardinality, no constraint is needed as it is the default case

- (2) For a relationship with 1:1 cardinality, define the constraint using the following pseudo-code:

  Table {CONSTRAINT *foreign-key* UNIQUE}

  For instance, for the 1:1 relationship between PROFESSOR and OFFICE, the constraint is

  PROFESSOR {CONSTRAINT *OfficeID* UNIQUE}

Defining Constraints (cont'd)

- (3) For a relation with 1 modality, no constraint is needed as it is the default case. Nevertheless, the following pseudo-code can be used to make the default explicit:

  Table {CONSTRAINT MANDATORY TO} The-Other-Table

  For instance, for the 1:1 relationship between PROFESSOR and OFFICE, the following constraint imposes 1 modality to the OFFICE table:

  OFFICE {CONSTRAINT MANDATORY TO} PROFESSOR

- (4) For a relation with 0 modality, define the constraint using the following pseudo-code:

  Table {CONSTRAINT OPTIONAL TO} The-Other-Table

  For instance, for the 1:1 relationship between PROFESSOR and OFFICE, the following constraint imposes 0 modality to the PROFESSOR table:

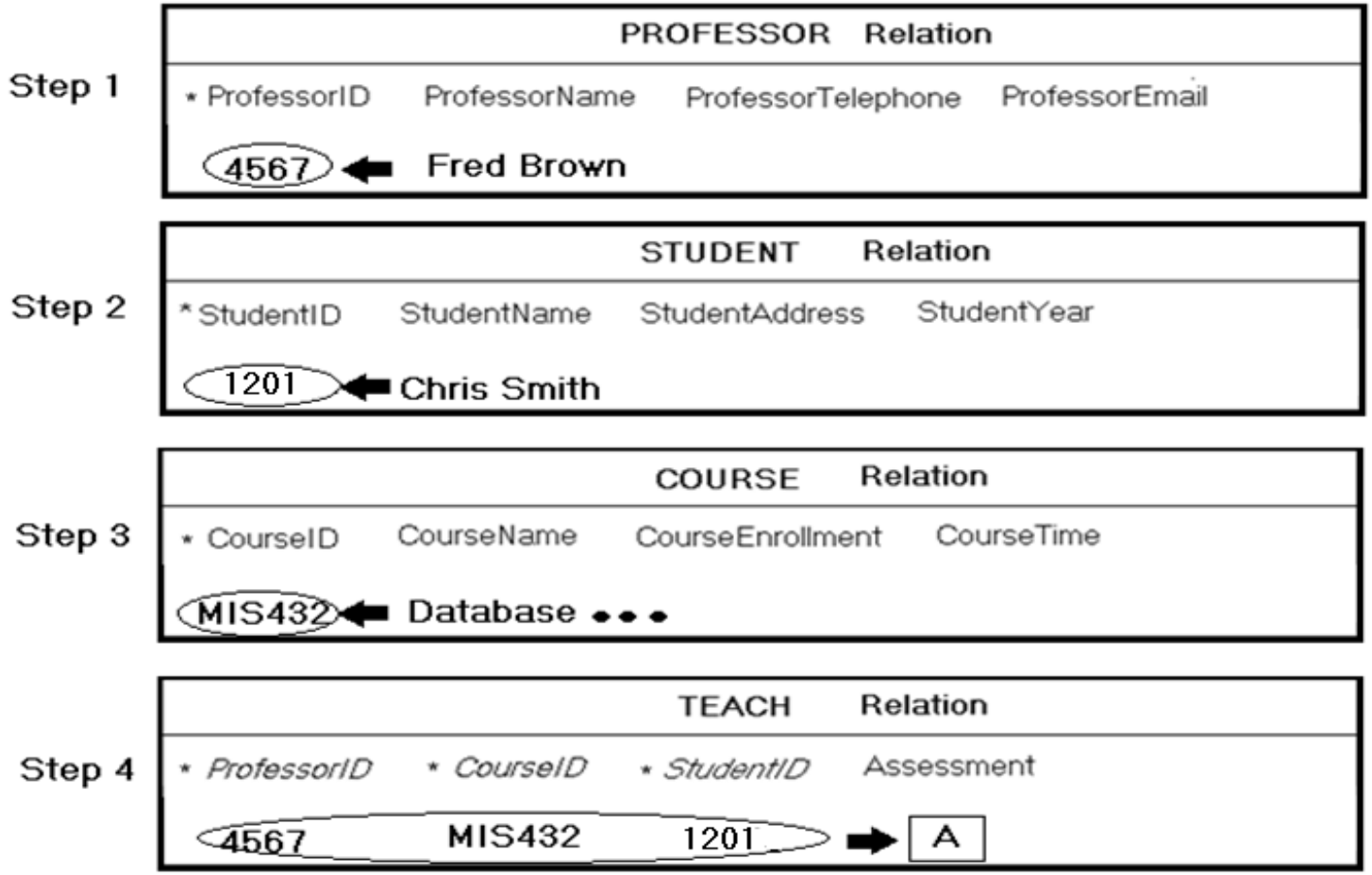
  PROFESSOR {CONSTRAINT OPTIONAL TO} OFFICE

Data Retrieval from Relational Database

- Simple cases: Data retrieval from a single table each time - Trace primary keys and foreign keys to find needed data
- Highly integrated data retrieval – Join multiple tables using keys and foreign keys of the tables

Example: Simple data retrieval

Find the assessment made by Professor Fred Brown for student Chris Smith in the Database Design & Implementation course.



Join Tables

List course ID, course title, course enrollment, course time for each course along with its classroom's data including classroom ID, classroom type, classroom size.

It is impossible to retrieve the needed data using the simple data retrieval approach

CLASSROOM Relation			
* ClassroomID	ClassroomLocation	ClassroomType	ClassroomSize

COURSE Relation				
* CourseID	CourseName	CourseEnrollment	CourseTime	*ClassroomID*

Merged Table

COURSE-CLASSROOM

CourseID	CourseName	CourseEnrollment	CourseTime	**ClassroomID**	ClassroomLocation	ClassroomType	ClassroomSize
MIS212	Programming	40	MWF 9-10	T-001	T-Building	Comp-Lab	50
MIS432	Database	30	MWF 11-12	T-001	T-Building	Comp-Lab	50
ACT211	Accounting I	40	TR 10-11	L-213	L-Building	Lecture	60
FIN312	Finance	35	MWF 9-10	L-213	L-Buldinbg	Lecture	60
MGT211	Organization	38	TR 10-11	K-123	K-Bulding	Case-Room	40

- The merged table contains redundant data
- The merged table provides data integration

Referential Integrity

- As the primary key in the 1-side table is the foreign key in the M-side table, there are potential violations of data integrity
- For example, when the user deletes a classroom records but the classroom has already been assigned to several courses, what should the database do?
- The DBMS controls the data integrity based on different rules

Delete Rule:

- (1) **Restrict –** If there is a record in another table that is relating to the record to be deleted, the deletion is prohibited

CLASSROOM Relation			
* ClassroomID	ClassroomLocation	ClassroomType	ClassroomSize

COURSE Relation				
* CourseID	CourseName	CourseEnrollment	CourseTime	*ClassroomID*

- A classroom record is allowed to be deleted only when it has not been assigned to any course

Delete Rule:

- (2) **Cascade –** All records in another table that are relating to the record to be deleted will be deleted as well

CLASSROOM Relation			
* ClassroomID	ClassroomLocation	ClassroomType	ClassroomSize

COURSE Relation				
* CourseID	CourseName	CourseEnrollment	CourseTime	*ClassroomID*

- If one deletes a classroom record, all records of the courses which use the classroom will be deleted automatically

Delete Rule:

- (3) **Set-to-Null** - All records in another table that are relating to the record to be deleted will be updated by setting the FK attribute to NULL

CLASSROOM Relation			
* ClassroomID	ClassroomLocation	ClassroomType	ClassroomSize

COURSE Relation				
* CourseID	CourseName	CourseEnrollment	CourseTime	*ClassroomID*

- If a classroom record is deleted, all courses that have been assigned to this classroom will be changed to “null” that means “no classroom”